Differential Equations and Linear Algebra - Set #8

7.4

8b
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> with(DEtools):
> phase portrait(2*diff(Q(t),t,t)+4*diff(Q(t),t),t)+20*Q(t)
  = 10*exp(-t), Q(t), t=0..10, [[Q(0)=0, D(Q)(0)=0]],
  step size = .05, arrows=NONE);
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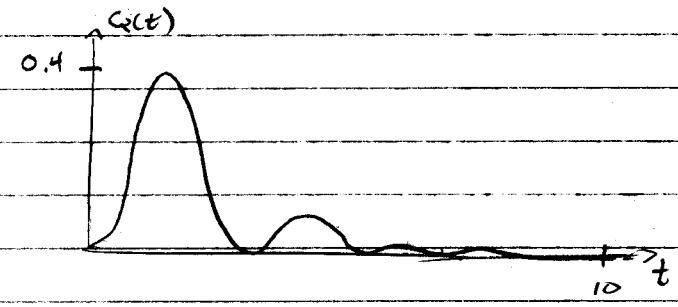

6.3

2c By inspection, $(x^4-x^2)-(x^4-x^2-1)=1=(x^4-x^2+1)-(x^4-x^2)$
so, solving for x^4-x^2+1 for instance,
$$x^4-x^2+1=2(x^4-x^2)-(x^4-x^2-1)$$

6.5

2p $y(x)=e^{rx}$ gives $r^4-2r^2-3=0$ so $r^2=(2\pm\sqrt{4+12})/2=3,-1$
so $r=\pm\sqrt{3},\pm i$
Therefore, $y(x)=C_1e^{\sqrt{3}x}+C_2e^{-\sqrt{3}x}+C_3\cos x+C_4\sin x$

4b $r^2+6r+9=0$ gives $r=-3,-3$ so $y(x)=(C_1+C_2x)e^{-3x}$.
The initial conditions give $y(x)=[2(1-e)-(2-3e)x]e^{-3(x-1)}$

4K. $r=0,0,0,0,0,0$ so $y(x)=C_1+C_2x+C_3x^2+C_4x^3+C_5x^4+C_6x^5$
The initial conditions give $y(x)=x^5/40$

8b contd

$$2\frac{d^2Q}{dt^2} + 4\frac{dQ}{dt} + 20Q = 10e^{-t}$$

Q_h: $2\frac{d^2Q}{dt^2} + 4\frac{dQ}{dt} + 20Q = 0$, $e^{rt} \rightarrow (2r^2 + 4r + 20)e^{rt} = 0$

$\therefore 2r^2 + 4r + 20 = 0 \rightarrow r = -1 \pm 3i \rightarrow Q_h = e^{-t}(C_1 \sin 3t + C_2 \cos 3t)$

Q_p: Let $Q_p = Ae^{-t}$; $2Ae^{-t} - 4Ae^{-t} + 20Ae^{-t} = 10e^{-t} \rightarrow A = \frac{5}{9}$

$\therefore Q(t) = Q_h + Q_p = \boxed{e^{-t}\left(\frac{5}{9} + C_1 \sin 3t + C_2 \cos 3t\right)}$

$\lim_{t \to \infty} Q(t) = 0$